

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Designed and implemented **3D game simulations** and environments using **Panda3D** for AI training applications, focusing on performance optimization for compatibility across various hardware.
- Developed and maintained **Python and C++ codebases** integrated with **Panda3D**, contributing to the core simulation logic and feature development for AI training scenarios.
- Collaborated with a multidisciplinary team to translate complex AI training requirements into actionable technical specifications and software solutions, ensuring alignment with project goals.
- Contributed to version control workflows using **GitHub** for code management, including branching strategies and pull request reviews to maintain code integrity and facilitate team collaboration.
- Documented architectural decisions, code implementations, and development processes, ensuring clarity and supporting collaborative efforts within the remote team.
- Participated in code reviews and team meetings, providing and receiving constructive feedback to enhance code quality and project outcomes.

SKILLS

Game Development – Panda3D, 3D graphics pipelines, simulation logic, AI training environments, asset optimization.

Programming Languages – Python, C++.

Version Control – GitHub, Git branching, pull requests.

Software Development – Agile methodologies, debugging, cross-platform development, performance optimization.

Documentation & Communication – Technical documentation, written and verbal communication, collaborative development.

PROJECTS

AI Training Simulation Environment (HCL Technologies)

- Developed a modular **Panda3D simulation environment** to generate synthetic data for training machine learning models, incorporating dynamic environmental elements and agent behaviors.
- Implemented **Python scripts** for procedural generation of 3D assets and environmental layouts, reducing manual creation time by approximately 40% for diverse training scenarios.
- Integrated data logging mechanisms to capture simulation states and agent actions, generating datasets used for evaluating model performance in areas like pathfinding and decision-making.

Interactive 3D Visualization Tool (MSc Project)

- Built an interactive **3D visualization tool** using **Panda3D** to represent complex data structures, enabling stakeholders to explore relationships and patterns visually.
- Developed data import and parsing modules in **Python** to handle various data formats, populating the 3D scene with nodes and edges representing connections.
- Implemented user-controlled camera navigation and object interaction features to enhance the exploratory capabilities of the visualization.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)